PROGRAMMING ON PYTHON 2

Skyway airlines Project Presentation

TEAM ALGEBRA

Ismail Yunussov
Assel Aidarkhankyzy
Ilya Zaitsev

Project objective

Goal

create console app for airlines company menegment(manipulating flights, tickets and passengers)

programm realisation

We created this app using sql database with sqlite library, creating classes with OOP and file handling

Database sturcture

Main Goal

We created 3 tables in our database to implement our app. We made tables for flights, passengers and tickets with fields that stores basic infrmations about them. One of the important things is the tickets table thet relates with passenger and flights table

```
CREATE TABLE IF NOT EXISTS Flights (
    flight_id INTEGER PRIMARY KEY AUTOINCREMENT,
    flight_number TEXT UNIQUE NOT NULL,
    departure_city TEXT NOT NULL,
    destination_city TEXT NOT NULL,
    departure_datetime TEXT NOT NULL,
    arrival_datetime TEXT NOT NULL,
    aircraft_model TEXT NOT NULL,
    total_seats INTEGER NOT NULL,
    base_price REAL NOT NULL,
    status TEXT NOT NULL DEFAULT 'Scheduled'
```

```
CREATE TABLE IF NOT EXISTS Passengers (
    passenger_id INTEGER PRIMARY KEY AUTOINCREMEN
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    passport_number TEXT UNIQUE NOT NULL,
    email TEXT NOT NULL,
    phone TEXT NOT NULL,
    date_of_birth TEXT NOT NULL
```

```
REATE TABLE IF NOT EXISTS Tickets (
  ticket_id INTEGER PRIMARY KEY AUTOINCREMENT,
  flight_id INTEGER NOT NULL,
  passenger_id INTEGER NOT NULL,
  seat_number TEXT NOT NULL,
  ticket_type TEXT NOT NULL,
  ticket_price REAL NOT NULL,
  booking_datetime TEXT NOT NULL,
  ticket_status TEXT NOT NULL DEFAULT 'Booked',
  FOREIGN KEY (flight_id) REFERENCES Flights (flight_id),
  FOREIGN KEY (passenger_id) REFERENCES Passengers (passenger_id),
  UNIQUE (flight_id, seat_number)
```

Models.py and usage of OOP and regex

classes creation

to simplify the work with functions that operate flights, passengers and tickets we created the corresponding classes

regex

to check the right email addres and phone number entering for passenger info we created two functions that use regex to check info for right pattern

REFLECTIONS & GROWTH

Main.py file

main information

main.py is core file for our app that does all the app functionality

CRUD operations

We made functions for database manipulating suck as insert, update delete, showing records and search

statistics

We can watch main app statistics about revenue, table recors quantity and etc.

file handling

our app can import main information into a txt file with main statistics, overview and flight schedule

programm work examples

```
==============================
       SKYWAY AIRLINES SYSTEM
==============================
1.  Add Flight
2.  Add Passenger
3.  Book Ticket
4.  Show Flights
5.  Show Passengers
6.  Show Tickets
7.  Search
8.  Update Record
9.  Cancel Ticket
10. Delete Record
11. Show Statistics
12. Export Report
0.  Exit
==============================
```

MAIN MENU

```
--- SkyWay Airlines Statistics ---
Total Flights           : 22
Total Passengers        : 50
Active Booked Tickets   : 38
Cancelled Tickets       : 4
Total System Seats      : 2926
Seat Occupancy Rate     : 1.30%
Total Ticket Revenue    : 3,538,000.00 KZT
```

STATISTICS OUTPUT

```
Enter city name: Astana

Found 5 flight(s):
Flight DV663: Astana -> Tbilisi (
Flight SW539: Tbilisi -> Astana (
Flight KZ666: Astana -> Istanbul
Flight KC414: Astana -> Tbilisi (
Flight KC106: Karaganda -> Astana
```

SEARCH EXAMPLE

Team members contribution

Ismail Yunussov

ticket booking, delete records, statistics ant txt import

Ilya Zaitsev

Datatbase structure and records inserting and showing records and canceling tickets

Assel Aidarkhankyzy

OOP logic and regex implementation in models.py, searching records and update